Question 1:

(a) (Theory)

The purpose of a mixer is to shift the frequency of an input signal to a different frequency (usually an intermediate frequency) to simplify signal processing and improve selectivity.

(b) (Theory)

A mixer produces two output frequencies:
- f_IF1 = f_input - f_LO
- f_IF2 = f_input + f_LO

The mathematical basis comes from the trigonometric identity:
$\sin(A)\sin(B) = 1/2\,[\cos(A - B) - \cos(A + B)]$

(c) (Computed example)

The two output frequencies produced by the mixer are:
- f_IF1 = 900 MHz - 860 MHz = 40 MHz
- f_IF2 = 900 MHz + 860 MHz = 1760 MHz

The lower frequency (40 MHz) is typically chosen as the IF because it is easier to filter and process at lower frequencies.

(d) (Theory)

The image frequency is an unwanted signal at a frequency that can also mix with the LO to produce the same IF.

(e) (Application of Theory)

For an IF of 40 MHz:
- f_image = f_LO + f_IF = 860 MHz + 40 MHz = 900 MHz

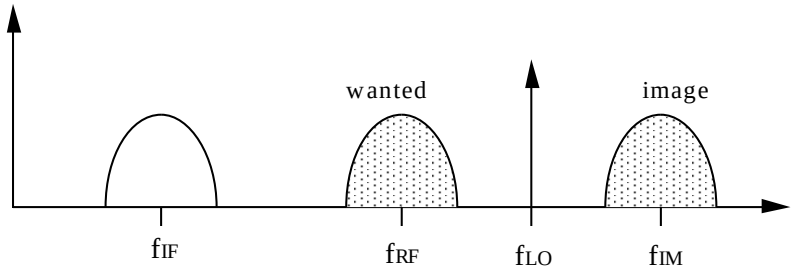
The image frequency must be filtered out to avoid interference.

(f) (Theory)

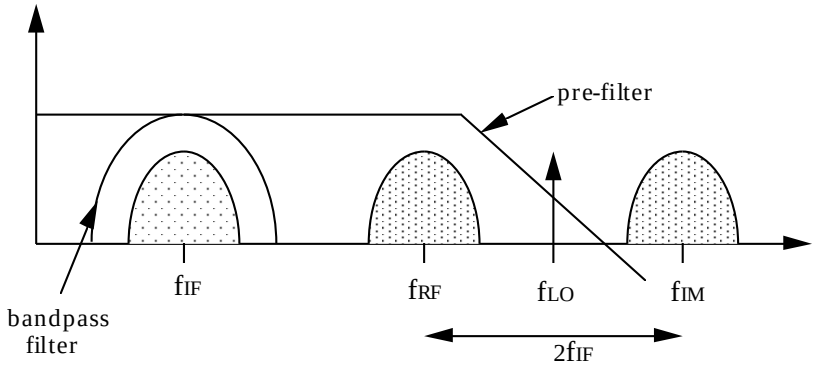
A mixer is a key component in a superheterodyne receiver. It combines the incoming RF signal with the local oscillator (LO) signal to produce an intermediate frequency (IF). This down-conversion simplifies the design of the bandpass filter needed for channel selection.

The tradeoff for the selection of the IF frequency is related to the image signal.

• Image signals: An image channel (f_{IM}) is also converted to the intermediate frequency, where $f_{IF} = f_{IM} - f_{LO}$



This image signal must be suppressed by filtering before downconversion. The pre-filter can be a simple low pass filter. The pre-filter should pass RF signals that are at $f_{LO}-f_{IF}$, but reject image signals at $f_{LO}+f_{IF}$. The design of the pre-filter is thus eased if f_{IF} is fairly high.



The bandpass filter that selects the required channel is centred at f_{IF}. The design of this filter is eased if f_{IF} is fairly low. Thus there is a trade-off between these two requirements.

Question 2:

(a) (Theory)

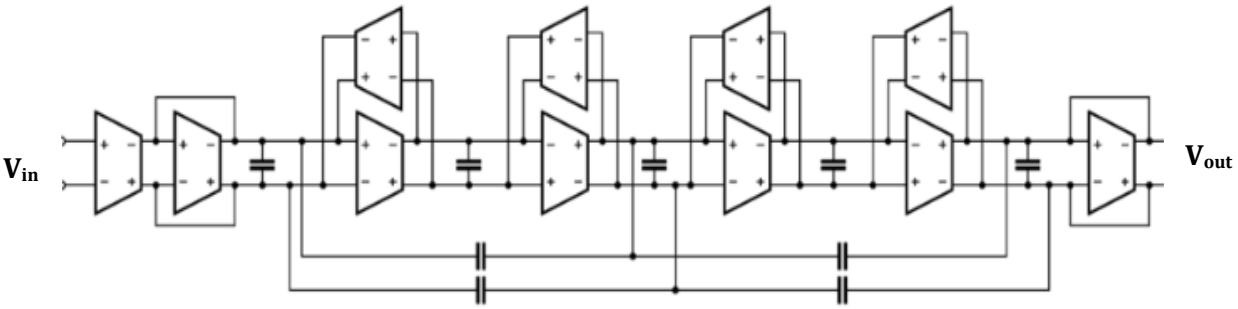
A biquad filter is a second-order digital or analog filter that can be represented in the following general form:

$H(s) = (b0 + b1 * s + b2 * s^2) / (a0 + a1 * s + a2 * s^2)$

(b) (Application of theory)

A linear phase response ensures that all frequency components are delayed equally, preserving the waveform shape. In audio processing, this is crucial to avoid phase distortion, which can alter the natural timbre and harmonic content of sound.

(c) (Practical example)



(d) (New theory. Application of theory)

$\dot{x}(t) = A\,x(t) + B\,u(t)$
$y(t) = C\,x(t) + D\,u(t)$

where:
$A = [0, 1; -1, -1]$
$B = [0; 1]$
$C = [1, 0]$
$D = [0]$

The transfer function is:
$H(s) = C\,(sI - A)^{-1}\,B + D$
$H(s) = (s + 1) / (s^2 + s + 1)$

Looking at the limits at zero an infinity this is a lowpass filter

(e) (Application of theory. Computed example)

$H(s) = 1 / (L C s^2 + (1 / C) s + 1)$, which in a more general form can be written as

$H(s) = \omega_c^2 / (s^2 + \sqrt{2}\, \omega_c s + \omega_c^2)$

Hence: $L = 1 / (\omega_c \sqrt{2})$; $C = \sqrt{2} / \omega_c$

For a 1 MHz Butterworth low-pass filter:
- $\omega_c = 2\pi \times 10^6$
- $L = 1 / (2\pi \times 10^6 \times \sqrt{2})$
- $C = \sqrt{2} / (2\pi \times 10^6)$

If students don't remember the general transfer function this can just be done calculating the gain.

Substituting values:

$L \approx 113$ nH

$C \approx 225$ nF

(f) (Application of theory. Computed example)

The simulated inductance using a gyrator is:

$L_sim = C_gyr / g_m^2$

where $g_m = 100\ \mu S = 100 \times 10^{-6}$ S

L_sim should match the calculated inductor value from part (e):

113 nH $= C_gyr \times (100 \times 10^{-6})^2$

$C_gyr = 113 \times 10^{-9} \times 10^{-8}$

$C_gyr \approx 1.13$ fF

(g) (Application of theory)

The cutoff frequency for a gyrator-based filter is:

$\omega_c = g_m / \sqrt{(C_gyr \times C)}$

If the capacitor values are fixed:

- Increasing transconductance → Increases the cutoff frequency

- Decreasing transconductance → Decreases the cutoff frequency

If the cutoff frequency is too high:

- Reduce g_m to lower the cutoff frequency

- Alternatively, increase the capacitor values to compensate for increased transconductance

If the cutoff frequency is too low:

- Increase g_m to raise the cutoff frequency

- Alternatively, decrease the capacitor values

Question 3:

(a) (Application of theory)

(b) (Application of theory)

Any non linear function can be represented in a polynomial form. In a differential configuration the even terms of the polynomial disappear. Considering the distortion is represented by the non linear terms of the polynomial, and those terms reduce (as a factorial) with the order, linearity increases because those terms get reduced.

(c) (Application of theory)

The VCM term means that that is not a large signal transconductor (i.e. the transconductance will be input dependent). Furthermore the added term that is technology dependent could lead to very significant variations in the operatin point post fabrication.

(d) (Application of theory)

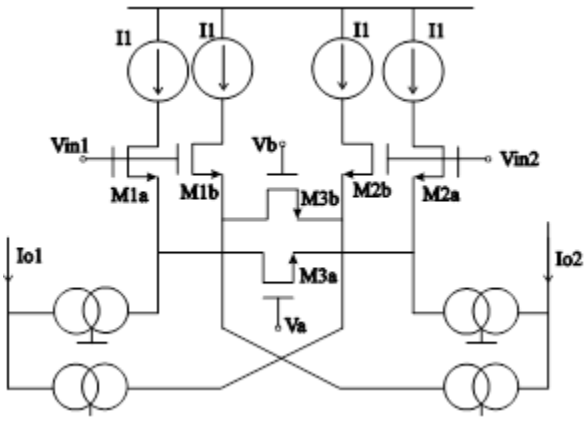
Replicate the circuit, connect the inputs, crosscouple the outputs, and connect a different V_G term to the two replicas off the circuit. The final transconductance will be:

$$I_{out}= \beta[\exp (cV_{G1})-\exp(cV_G2)]V_d;$$

(e) (New theory)

Predistorting the inputs with a ln function could help compensate for the exponential.

(f) (Theory)



$$Iout = 2\beta(Va - Vb)\ Vin$$

Question 4:

(a) (New theory)

$I_{out}=\beta(Vin-Vt)Vbias$

(b) (New theory)

$Gm=\beta Vbias$

(c)(New theory)

$I_{out}=\beta((Vin_{+}-Vin_{-})Vbias$

(d) (New theory)

$Gm=\beta((Vin_{+}-Vin_{-})$

(e) (New theory)

Just add an capacitor across the differential output, ideally split in two and flip plates.

(f) (New theory)

It is saturated: ground